

The Hong Kong University of Science and Technology
IEDA 1250, Summer 2026
Final Exam

Student ID _____

Name: _____

Please read the following instructions carefully.

- You have **120** minutes to complete this exam. This question booklet contains **7** questions for a total of 120 points. Check to see if any pages are missing.
- Read the instructions for individual questions carefully before answering the questions.
- **Please write your answers in the answer book.** You can use this question booklet as scratch paper.

Question	Points	Score
1	20	
2	20	
3	20	
4	20	
5	20	
6	10	
7	10	
Total	120	

1. (20 points) Consider the zero-sum game having the following payoff table for Player 1:

Strategy		Player 2	
		S1	S2
Player 1	S1	3	−2
	S2	−1	2

- (a) Use the graphical procedure to determine the value of the game and the optimal mixed strategy for each player.
- (b) Check your answer by (i) constructing Player 2's payoff table and (ii) applying the graphical procedure again to the resulting table.

2. (20 points) Consider the following zero-sum two-player game. The entries in the table are the payoffs to Player A.

	B1	B2	B3	B4	B5	B6
A1	1	10	2	3	11	12
A2	4	8	5	6	9	10
A3	7	5	8	9	6	7
A4	10	1	11	12	2	3
A5	3	7	4	5	8	9
A6	6	4	7	8	5	6

Player A chooses one of the six rows, and Player B chooses one of the six columns. Player A wants to maximize the expected payoff, while Player B wants to minimize the expected payoff.

- Find the Nash equilibrium in this game and report the value of the game.
- Write Player A's linear program for the *full* payoff table.
- The following linear program is associated with Player B. Let q_j be the probability that Player B chooses column j , for $j = 1, \dots, 6$, and let v_B represent the expected loss of Player B:

$$\begin{aligned}
 \min \quad & v_B \\
 \text{s.t.} \quad & q_1 + 10q_2 + 2q_3 + 3q_4 + 11q_5 + 12q_6 \leq v_B, \\
 & 4q_1 + 8q_2 + 5q_3 + 6q_4 + 9q_5 + 10q_6 \leq v_B, \\
 & 7q_1 + 5q_2 + 8q_3 + 9q_4 + 6q_5 + 7q_6 \leq v_B, \\
 & 10q_1 + q_2 + 11q_3 + 12q_4 + 2q_5 + 3q_6 \leq v_B, \\
 & 3q_1 + 7q_2 + 4q_3 + 5q_4 + 8q_5 + 9q_6 \leq v_B, \\
 & 6q_1 + 4q_2 + 7q_3 + 8q_4 + 5q_5 + 6q_6 \leq v_B, \\
 & q_1 + q_2 + q_3 + q_4 + q_5 + q_6 = 1, \\
 & q_1, q_2, q_3, q_4, q_5, q_6 \geq 0, \\
 & v_B \text{ is unrestricted in sign.}
 \end{aligned}$$

Show that Player A's LP and Player B's LP are primal and dual LPs. That is, starting from Player A's LP in part (b), you need to derive its dual and show that the dual is identical to the given one.

3. (20 points) A college student has 7 days remaining before final examinations begin in her four courses, and she wants to allocate this study time as effectively as possible. She needs at least 1 day on each course, and she likes to concentrate on just one course each day, so she wants to allocate 1, 2, 3, or 4 days to each course. Having recently taken an optimization course, she decides to use dynamic programming to make these allocations to maximize the total grade points to be obtained from the four courses. She estimates that the alternative allocations for each course would yield the number of grade points shown in the following table:

Study Days	Course 1	Course 2	Course 3	Course 4
1	3	5	2	6
2	5	5	4	7
3	6	6	7	9
4	7	9	8	9

Solve this problem by dynamic programming. Show the backward induction calculations. You do not need to write down the recursive equations.

4. (20 points) A food-truck owner must decide where to operate during each of the next 5 days. Each day, she can choose one of three locations: A , B , and C . The estimated profit from operating at each location on each day is shown below:

Location	Day 1	Day 2	Day 3	Day 4	Day 5
A	8	6	7	9	6
B	5	9	6	7	10
C	6	7	10	5	8

However, changing locations from one day to the next incurs a travel and setup cost. The cost of moving from one location to another is:

From/To	A	B	C
A	0	3	4
B	3	0	2
C	4	2	0

At the beginning of Day 1, the truck is already located at A . Therefore, if she operates at location A on Day 1, there is no travel cost. If she operates at location B or C on Day 1, she must pay the corresponding travel cost from A . There is one additional restriction: she may not operate at the same location for three consecutive days (i.e., she may operate at the same location for at most two consecutive days). The goal is to choose the operating location for each of the 5 days to maximize total profit minus travel costs.

Suppose we define the state before day t as $S_t = (L, r)$, where $L \in \{A, B, C\}$ is the location where the truck operated most recently before day t , and $r \in \{0, 1, 2\}$ is the number of consecutive days, immediately before day t , that the truck has operated at location L . For Day 1, the initial state is $S_1 = (A, 0)$.

Solve this problem using backward induction. Find the optimal sequence of locations and the maximum total net profit (you do not need to write down recursive equations, but you must show the dynamic programming calculations clearly).

5. (20 points) Your system consists of a single CPU with finite buffer capacity. Jobs arrive according to a Poisson process with rate λ jobs/sec. The job sizes are exponentially distributed with mean $1/\mu$ seconds. Jobs are serviced in FCFS order. Let $K - 1$ denote the maximum number of jobs that your system can hold in the queue. Thus, including the job in service, there are a maximum of K jobs in the system at any one time. If a job arrives when there are already K jobs in the system, then the arriving job is rejected.

Your Defense Advanced Research Projects Agency (DARPA) proposal requires that you reduce the probability of rejecting jobs in your system. To do this you could either ask for money to double the system capacity, or, alternatively, you could ask for money to double the speed of the CPU so that jobs get processed faster, thereby lowering the probability that there are K jobs in the system. Assuming both proposals have the same cost, which do you choose? (Asking for both is too expensive.)

Suppose that $K = 4$, $\lambda = 2$, and $\mu = 4$, then these are the specific questions you should answer:

- (a) Draw the CTMC.
- (b) Derive the stationary probabilities π_n for $n = 0, \dots, 4$.
- (c) What is the utilization of the system, i.e., the fraction of time that the server is busy?
- (d) What is the fraction of jobs turned away (loss probability)? Use the word PASTA in your explanation.
- (e) What is $\mathbb{E}[N]$, i.e., the expected number of jobs in the system?
- (f) What is $\mathbb{E}[T]$ for only those jobs that enter the system. Hint: Here you need to use *the effective arrival rate* $\lambda_e := \sum_{n=0}^K \lambda_n \pi_n$, where λ_n is the arrival rate to the system with n jobs.
- (g) Which would have the greater effect on lowering loss probability: doubling the system capacity from $K = 4$ to $K = 8$ or doubling the CPU speed from $\mu = 4$ to $\mu = 8$?
- (h) Answer the same question as (g) except now $K = 4$, $\lambda = 3$, and $\mu = 4$.
- (i) Explain intuitively why (g) and (h) resulted in different answers.

6. (10 points) Suppose your computer center currently consists of a single server of speed μ . Jobs arrive according to a Poisson process with rate λ , and their service times are exponentially distributed.

Suppose the current response time is considered intolerable by the users. A second, faster server, running at speed $\alpha\mu$ (for $\alpha > 1$), is purchased and added to the system. With the second server, the system becomes a variant of the M/M/2 queue where the service rates of the two processors are not identical. Denote the service rate of the first processor by $\mu_1 = \mu$ and the service rate of the second processor by $\mu_2 = \alpha\mu$, where $\mu_1 < \mu_2$. In such system, the rule is that when both servers are idle, the faster server is scheduled for service before the slower one. Define the utilization, $\rho (< 1)$, for this system to be $\rho = \lambda/(\mu_1 + \mu_2)$.

Suppose you are told that the mean response time $\mathbb{E}[T]$ in the system is given by

$$\mathbb{E}[T] = \frac{1}{\lambda A (1 - \rho)^2},$$

where

$$A = \frac{\mu_1 \mu_2 (1 + 2\rho)}{\lambda(\lambda + \mu_1)} + \frac{1}{1 - \rho}.$$

Then determine the mean number of jobs $\mathbb{E}[N]$ in the system (you do not need to simplify your answer).

7. (10 points) In Question 6, a hotshot consultant walks in and makes the radical proposal of disconnecting the original server entirely (i.e., simply letting the faster server run by itself). Clearly this makes sense with respect to power, but the consultant claims this is also a win for $\mathbb{E}[N]$. For a consulting fee of \$400/hr, what is the consultant thinking? Come up with an instance, with specific values for λ , μ_2 , and μ_1 for which the consultant is right. Also, explain intuitively what is happening. [If you find this problem interesting, you can think about a general criterion under which the consultant would be right... Throwing away servers is fun!]